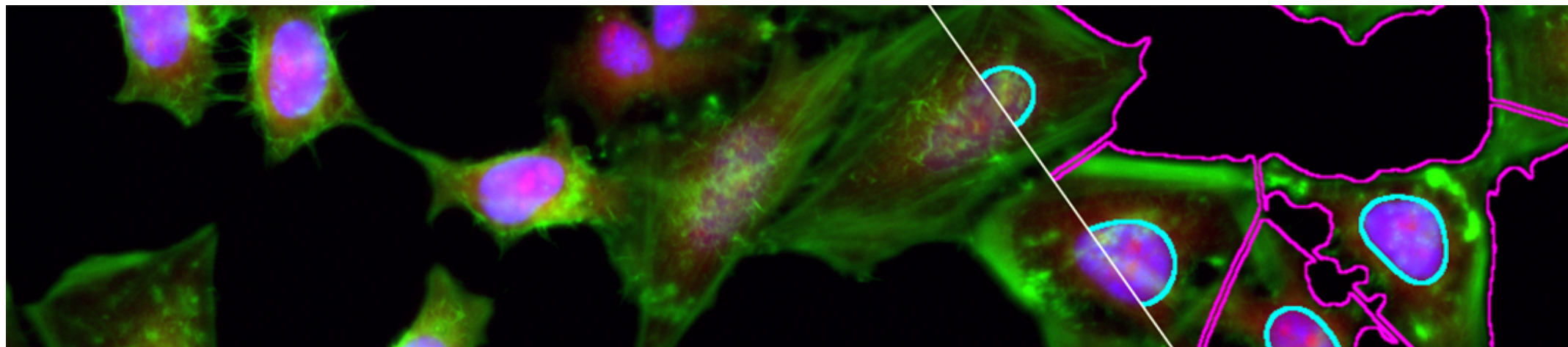


[5. Segmentation, From Intensities To](#)
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5.1.1. Description & Outline



1. Digital Images
2. Of Colors & Dimensions
3. Resizing, Interpolation & Mathematical Operations
4. Filtering

5. Segmentation, from Intensities to Objects

6. Regions of Interest & Results
 7. Color in Depth, Projections & Reslicing
- Extra. ImageJ Macro Language Programming Primer

Week Overview

- Section 1: Introduction
- Section 2: Segmentation
- Section 3: Morphological Operations
- Section 4: Analyze Particles
- Section 5: Conclusion

Description

This week tackles the biggest challenge of image processing and analysis: Segmentation. That is, converting the pixel intensities into labels or regions that a computer can subsequently extract data from. We will cover several segmentation techniques, most of which will result in what are called binary masks. These binary masks can be further processed using a particular set of operations dedicated to black and white images, Morphological Operations. Finally, after these, we will learn how to extract per-object statistics using Fiji and the 'Analyse Particles' command.

By the end of this week, you will be able to describe how a computer can "understand" what an object is and describe at least 3 forms of image segmentation. You will be able to apply image segmentation techniques on appropriate images. You will be able to recognize and apply morphological operations on binary images. Finally you will know how to apply particle (object) analysis on a segmentation result, and how to present these results in a human readable format through Fiji's Results tables.

Download Images

Please find below, a link to download some images that we'll use during the course.

[Week 5 images](#)

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